

Homework B

Klik en typ de subtitel

Enmin Lin, Kaixi Yao

april 2026

1 Q1



The homework defines the prior $p(x) = 0.5 \mathcal{N}(x \mid -1.0, 1) + 0.5 \mathcal{N}(x \mid 1.0, 0)$.

I keep the statement as written. I interpret the second term as a degenerate Gaussian, that is, a point mass at $x = 1$. I sample 10000 prior values with random seed 42. These prior samples have mean -0.00800 and standard deviation 1.22944 .

No extra normalization is needed. The mixture weights already sum to one.

The measurement model is $z = 2x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.3)$, $z = 0.5$.

I update the prior with Bayes' rule: $p(x \mid z) \propto p(z \mid x) p(x)$.

As Fig. 1 shows, the prior has a broad Gaussian component around $x = -1$ and a spike at $x = 1$. The posterior shifts toward values that explain $z = 0.5$, so the continuous part moves toward $x \approx 0.25$. The point mass at $x = 1$ remains, but its weight decreases.

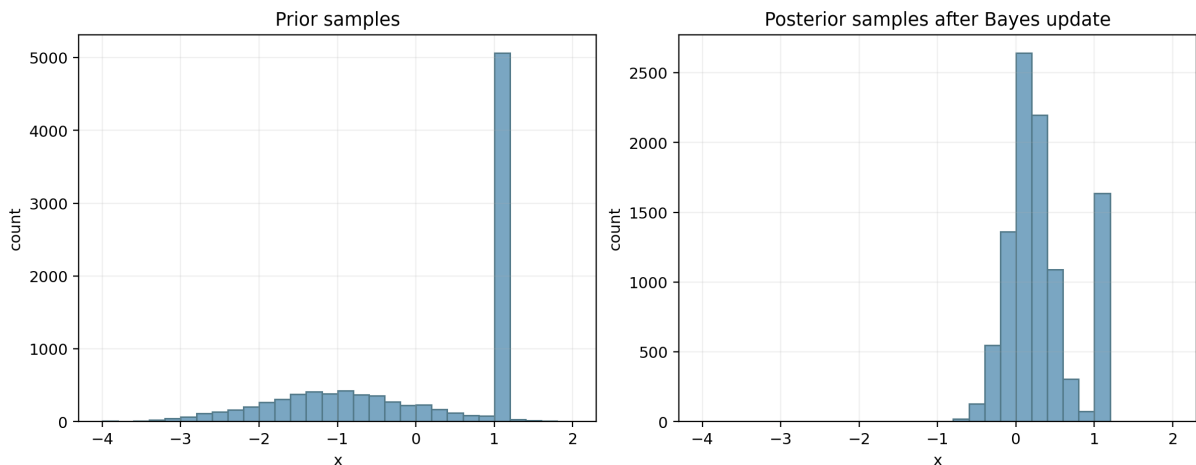


Fig. 1. Prior and posterior sample histograms for Homework B1.

The posterior sample mean is 0.30263 , and the posterior standard deviation is 0.38992 . The resulting distribution is not a single Gaussian. It is a mixed distribution. More precisely, it is a Gaussian mixture with one degenerate component.

2 Q2

We are given a 6-DOF robot at joint position q , with Jacobian $J(q)$. The end effector has a plane passing through it. The plane is described by two vectors u and v , which lie inside the plane. The joint velocities are limited by $\|\dot{q}\|_2 \leq 1$.

The end-effector translational velocity is given by $\dot{p} = J_p(q)\dot{q}$ where $J_p(q) \in \mathbb{R}^{3 \times 6}$ is the translational part of the Jacobian. Since the question asks about velocities inside a plane, we focus on the end-effector linear velocity $\dot{p}$, not the angular velocity.

2.1 Step 1: Build an Orthonormal Basis for the Plane

The vectors u and v lie on the plane, but they are not orthogonal. So we first convert them into an orthonormal basis using Gram-Schmidt.

First define $e_1 = \frac{u}{\|u\|}$. Then remove from v the part that points in the e_1 direction: $v_{\perp} = v - e_1(e_1^T v)$. Normalize the result: $e_2 = \frac{v_{\perp}}{\|v_{\perp}\|}$. Now define $E = [e_1 \ e_2] \in \mathbb{R}^{3 \times 2}$. The columns of E form an orthonormal basis for the plane. Any velocity inside the plane can now be written as $\dot{p}_{\perp} = Ey$, where $y \in \mathbb{R}^2$ is the coordinate vector of the velocity inside the plane. Because e_1 and e_2 are orthonormal, $\|\dot{p}_{\perp}\| = \|y\|$. So the magnitude of the real 3D velocity is equal to the magnitude of its 2D coordinate vector y .

2.2 Shape of the Reachable Velocity Set

The translational end-effector velocity is $\dot{p} = J_p \dot{q}$. Project this velocity into the plane coordinates: $y = E^T \dot{p}$. Substitute $\dot{p} = J_p \dot{q}$: $y = E^T J_p \dot{q}$. Define $A = E^T J_p$. Then $y = A \dot{q}$. The joint velocity constraint is $\|\dot{q}\|_2 \leq 1$. Therefore, the set of all reachable in-plane velocities is $V_{\perp} = \{A \dot{q} : \|\dot{q}\|_2 \leq 1\}$. This means that we take the unit ball in joint velocity space and map it into the 2D plane using the matrix A . To understand the shape, use the singular value decomposition: $A = U \Sigma V^T$. Then $A \dot{q} = U \Sigma V^T \dot{q}$. Each part has a simple geometric meaning:

- V^T rotates the input ball. A ball remains a ball after rotation.
- Σ scales the ball differently along different directions.
- U rotates the result in the output plane.

A circle or ball that is stretched differently along different directions becomes an ellipse or ellipsoid.

Therefore, the reachable velocity set inside the plane is an ellipse:

The set of all reachable velocities inside the plane is an ellipse.

If the robot is at a singular configuration, this ellipse may degenerate into a line segment or a point.

2.3 Maximum Velocity Inside the Plane

We want the largest possible in-plane velocity magnitude: $v_{max} = \max_{\|\dot{q}\|_2 \leq 1} \|A \dot{q}\|$

This is exactly the largest singular value problem. If $A = U \Sigma V^T$

And the singular values are $\sigma_1 \geq \sigma_2 \geq 0$

Then the maximum velocity is the major semi-axis length of the ellipse: $v_{max} = \sigma_{max}(A)$

Since $A = E^T J_p$

We obtain $v_{max} = \sigma_{max}(E^T J_p)$ Equivalently, using eigenvalues: $v_{max} = \sqrt{\lambda_{max}(A A^T)}$

Substituting $A = E_T J_p$:
$$v_{max} = \sqrt{\lambda_{max}(E^T J_p J_p^T E)}$$

This uses only linear algebra operations, such as Gram-Schmidt, SVD, and eigenvalue decomposition. It does not require numerical optimization.

3 Q3

We use the standard 10-dimensional inertial parameter vector

$$\mu = [m \quad mc_x \quad mc_y \quad mc_z \quad I_{xx} \quad I_{xy} \quad I_{xz} \quad I_{yy} \quad I_{yz} \quad I_{zz}]^T.$$

Here:

- m is the mass of the link.
- $c = [c_x, c_y, c_z]^T$ is the position of the center of mass expressed in the chosen frame.
- I is the 3×3 inertia matrix about the origin of the chosen frame.
- mc is the first moment of mass.

The inertia matrix is
$$I = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}.$$

The notation $[c]_{\times}$ means the skew-symmetric matrix corresponding to the cross product with

$$c: [c]_{\times} = \begin{bmatrix} 0 & -c_z & c_y \\ c_z & 0 & -c_x \\ -c_y & c_x & 0 \end{bmatrix}.$$

For any vector r , $[c]_{\times} r = c \times r$.

3.1 Transforming μ^a to μ^b

Let ${}^aT_b = ({}^aR_b, {}^ap_b)$

be the rigid-body transform from frame $\{b\}$ to frame $\{a\}$. This means that a point satisfies $x^a = {}^aR_b x^b + {}^ap_b$

First, recover the mass, center of mass, and inertia matrix from μ_a :
$$m, c^a = \frac{1}{m} \begin{bmatrix} mc_x^a \\ mc_y^a \\ mc_z^a \end{bmatrix}, I^a.$$

The mass does not change: $m^b = m^a = m$.

The center of mass in frame $\{b\}$ is $c^b = ({}^aR_b)^T (c^a - {}^ap_b)$



For the inertia matrix, we should not simply rotate I^a , because I^a is the inertia about the origin of frame $\{a\}$, not necessarily about the center of mass. We first use the parallel-axis theorem to move the inertia to the center of mass: $I_c^a = I^a - m[c^a]_\times^T [c^a]_\times$.

Then we rotate this center-of-mass inertia into frame $\{b\}$: $I_c^b = ({}^aR_b)^T I_c^a {}^aR_b$.

Finally, we use the parallel-axis theorem again to express the inertia about the origin of frame $\{b\}$: $I^b = I_c^b + m[c^b]_\times^T [c^b]_\times$.

The transformed parameter vector is then $\mu_b = [m \quad mc_x^b \quad mc_y^b \quad mc_z^b \quad I_{xx}^b \quad I_{xy}^b \quad I_{xz}^b \quad I_{yy}^b \quad I_{yz}^b \quad I_{zz}^b]^T$.

In words: mass stays the same, the center of mass is transformed like a point, and the inertia matrix must be shifted to the center of mass, rotated, and shifted back to the new frame origin.

3.2 Physical Realizability

A mass parameterization μ is physically realizable if it can correspond to a real rigid body made of positive mass. Based on the lecture material, a real rigid body has positive mass and a valid inertia matrix about its center of mass.

Given $\mu = [m \quad mc_x \quad mc_y \quad mc_z \quad I_{xx} \quad I_{xy} \quad I_{xz} \quad I_{yy} \quad I_{yz} \quad I_{zz}]^T$,

first reconstruct the center of mass: $c = \frac{1}{m} \begin{bmatrix} mc_x \\ mc_y \\ mc_z \end{bmatrix}$.

Then reconstruct the inertia matrix about the current frame origin: $I = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}$.

Next, compute the inertia matrix about the center of mass using the parallel-axis theorem: $I_c = I - m[c]_\times^T [c]_\times$.

The parameterization is physically realizable if:

1. The mass is positive: $m > 0$.
2. The center-of-mass inertia matrix is symmetric: $I_c = I_c^T$.
3. The center-of-mass inertia matrix is positive definite: $I_c > 0$. This means the rotational kinetic energy is positive for every nonzero angular velocity.
4. The principal moments of inertia satisfy the triangle inequalities.

Let I_1 , I_2 , and I_3 be the eigenvalues of I_c , also called the principal moments of inertia. They should satisfy $I_1 + I_2 \geq I_3$, $I_1 + I_3 \geq I_2$, $I_2 + I_3 \geq I_1$.

These conditions express the fact that the inertia must come from a real distribution of positive mass. A matrix can be symmetric positive definite but still fail to represent a real rigid-body inertia if the principal moments violate these inequalities.

3.3 Rigidly Linked Bodies

Suppose body a has inertial parameters μ_a and body b has inertial parameters μ_b . The question statement appears to repeat μ_a for both bodies, but physically the second body should have its own parameters μ_b .

If the two bodies are rigidly linked, they form one composite body ab .

If both parameter vectors are already expressed in the same frame, then the composite inertial parameters are simply the sum: $\mu_{ab} = \mu_a + \mu_b$.

This works because mass, first moment of mass, and inertia matrix are additive: $m_{ab} = m_a + m_b$, $(mc)_{ab} = m_a c_a + m_b c_b$, $I_{ab} = I_a + I_b$.

Therefore, $\mu_{ab} = \begin{bmatrix} m_a + m_b \\ m_a c_a + m_b c_b \\ I_a + I_b \end{bmatrix}$, where the notation means that the corresponding entries of

the 10-dimensional parameter vectors are added. If μ_a and μ_b are expressed in different frames, we must first transform them into one common frame using the method from part (a).

Only after that can we add them: $\mu_{ab}^{\{o\}} = \mu_a^{\{o\}} + \mu_b^{\{o\}}$.

4 Q4

For all three cases, the contact is assumed to be frictionless and infinitely stiff. This is an ideal contact. In the lecture terminology, an ideal contact does not dissipate or store energy, so the contact wrench and object twist satisfy the reciprocity relation

$$w^T t = \tau^T \omega + F^T v = 0.$$

Here the twist and wrench are written as

$$t = [\omega_x \quad \omega_y \quad \omega_z \quad v_x \quad v_y \quad v_z]^T, w = [\tau_x \quad \tau_y \quad \tau_z \quad F_x \quad F_y \quad F_z]^T.$$

4.1 Peg insertion

Choose the task frame as follows:

- z_t is along the peg axis and insertion direction.
- x_t and y_t are lateral directions perpendicular to the peg axis.

The peg is cylindrical and is inserted into a round hole. The hole prevents lateral motion and tilting of the peg: $v_x = 0, v_y = 0, \omega_x = 0, \omega_y = 0$.

Because the contact is frictionless, the environment cannot exert a friction force along the insertion axis, and it cannot exert a frictional torque around the peg axis: $F_z = 0, \tau_z = 0$.

Therefore, the natural constraints are $v_x = 0, v_y = 0, \omega_x = 0, \omega_y = 0, F_z = 0, \tau_z = 0$

The artificial constraints are the complementary components: $F_x, F_y, \tau_x, \tau_y, v_z, \omega_z$

A consistent task-frame specification is:

Direction	Specification
x_t	Force control: $F_x = 0$
y_t	Force control: $F_y = 0$
z_t	Velocity control: $v_z = v_{\text{ins}}$
αx_t	Torque control: $\tau_x = 0$
αy_t	Torque control: $\tau_y = 0$
αz_t	Velocity control: $\omega_z = 0$

Here v_{ins} is the desired insertion velocity. Its sign depends on the chosen positive direction of z_t .

4.2 Flat Object on a Plane

Choose the task frame as follows:

- z_t is normal to the plane.
- x_t and y_t lie on the plane.

The plane is infinitely stiff, so the object cannot move through the plane: $v_z = 0$.

Because the bottom of the object is planar, the object also cannot tilt with respect to the support plane: $\omega_x = 0, \omega_y = 0$.

Because the contact is frictionless, the plane cannot generate tangential forces. It also cannot generate a frictional torque around the plane normal: $F_x = 0, F_y = 0, \tau_z = 0$.

Therefore, the natural constraints are $v_z = 0, \omega_x = 0, \omega_y = 0, F_x = 0, F_y = 0, \tau_z = 0$

The artificial constraints are the complementary components: $v_x, v_y, \omega_z, F_z, \tau_x, \tau_y$

A consistent task-frame specification is:

Direction	Specification
x_t	Velocity control: $v_x = v_{xd}$
y_t	Velocity control: $v_y = v_{yd}$

Direction	Specification
z_t	Force control: $F_z = F_{nd}$
αx_t	Torque control: $\tau_x = 0$
αy_t	Torque control: $\tau_y = 0$
αz_t	Velocity control: $\omega_z = \omega_{zd}$

If z_t points upward from the plane, then the normal support force usually satisfies $F_z > 0$.

Intuitively, the environment determines that the object cannot move down through the plane or tilt into it. The controller determines how the object slides and rotates in the plane, and how much normal contact force is applied.

4.3 Wedge on a Plane

Use the task frame shown in the figure:

- x_t lies in the plane and is perpendicular to the contact line.
- y_t lies along the contact line.
- z_t is normal to the plane.

The wedge contacts the plane along a straight line. Assume the task-frame origin lies on this contact line. A point on the contact line can then be written as $r = \begin{bmatrix} 0 \\ s \\ 0 \end{bmatrix}$.

The velocity of this point is $v(r) = v + \omega \times r$.

The normal component is $v_z(r) = v_z + \omega_x s$.

Since every point on the contact line must remain on the plane, this normal velocity must be zero for all values of s . Therefore, $v_z = 0, \omega_x = 0$.

So the environment naturally prevents the wedge from moving into the plane and from rotating around x_t .

Because the contact is frictionless, the contact force can only be normal to the plane. Thus the environment cannot generate tangential forces: $F_x = 0, F_y = 0$.

A normal force distributed along the contact line can generate a moment around x_t . To see this, take a normal force at a point on the line: $F = \begin{bmatrix} 0 \\ 0 \\ F_z \end{bmatrix}$.

The moment is $\tau = r \times F = \begin{bmatrix} 0 \\ s \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ F_z \end{bmatrix} = \begin{bmatrix} sF_z \\ 0 \\ 0 \end{bmatrix}$.

Therefore, the line contact can generate τ_x , but it cannot generate τ_y or τ_z : $\tau_y = 0, \tau_z = 0$.

Therefore, the natural constraints are $v_z = 0, \omega_x = 0, F_x = 0, F_y = 0, \tau_y = 0, \tau_z = 0$

The artificial constraints are the complementary components: $v_x, v_y, \omega_y, \omega_z, F_z, \tau_x$

A consistent task-frame specification is:

Direction	Specification
x_t	Velocity control: $v_x = v_{xd}$
y_t	Velocity control: $v_y = v_{yd}$
z_t	Force control: $F_z = F_{nd}$
αx_t	Torque control: $\tau_x = \tau_{xd}$
αy_t	Velocity control: $\omega_y = \omega_{yd}$
αz_t	Velocity control: $\omega_z = \omega_{zd}$

If z_t points upward from the plane, then the normal force usually satisfies $F_z > 0$.

The value of τ_x can be positive, negative, or zero. It depends on how the normal contact force is distributed along the contact line relative to the task-frame origin. If no particular pitching moment is desired, a common choice is $\tau_x = 0$.

4.4 Relationship Between Screw Twist and Wrench in Any Object-Attached Frame

For all three cases, contact is ideal. Therefore, the twist and wrench satisfy

$$w^T t = \tau^T \omega + F^T v = 0$$

This relation is independent of the frame in which the twist and wrench are expressed.

Let $\{t\}$ be the task frame and $\{o\}$ be any frame attached to the object. The twist and wrench transform as $t_o = \text{Ad}_T t_t, w_o = \text{Ad}_T^T w_t$.

Then $w_o^T t_o = (\text{Ad}_T^T w_t)^T (\text{Ad}_T t_t) = w_t^T t_t$. So if $w_t^T t_t = 0$, then also $w_o^T t_o = 0$.

Hence, in any object-attached frame, the admissible twist space and the admissible contact-wrench space remain reciprocal. The coordinate values change from frame to frame, but the physical relation between twist and wrench does not.